Triangle/Sine Converter for FORMANT LFOs

Synthesizers frequently require low-frequency sine waves for modulation purposes. However, the sub-audio range demands a precise circuit design—and here it is.

The sine signal is derived from the triangle-wave output of a FORMANT LFO. A simple triangle-to-sine converter—using just two diodes—was previously discussed in the context of the FORMANT VCO to round off the triangle waveform; for that purpose, two diodes were entirely sufficient.

The quality of such a converter can be improved by using more diodes; the circuit shown in Figure 1 employs ten diodes in a negative feedback configuration.

Since the circuit's performance depends heavily on the matching of diode characteristics, the diodes must be carefully selected for identical forward and reverse resistance values. With reasonably good selection, total harmonic distortion can easily be reduced to below 1%. As the circuit contains no frequency-dependent components, it is suitable for applications beyond the FORMANT LFO module.

Assembly and Calibration

The circuit can be used to provide a sine-wave output for the LFO module. The PCB (Figure 2) has been designed—in terms of dimensions and mounting holes—so that it can be attached to the FORMANT LFO board using 20–30 mm spacers. A suggested mounting arrangement is shown in Figure 3. The triangle-wave output of one of the three LFOs must now be connected to the input of the triangle-to-sine converter. A stranded wire connects the converter's output to the front panel, replacing the original triangle-wave output connection. No modifications to the LFO front panel itself are required. However, you may wish to mark the sine-wave output with an “S” or another type of dry-transfer letter or symbol.

For calibration—which requires a fully calibrated LFO module (or another triangle-wave voltage source)—start by setting both trimmers, P1 and P2, to their mid-position. An oscilloscope is required for precise adjustment of the operating point (Figure 4). Trimmer P2 is used to adjust the output voltage level.

Applications

The LFO module's sine wave output is particularly well-suited for generating vibrato (FM) and tremolo (AM) effects, which retain a smooth, rounded sound even in the 0.5–10 Hz range. The sine signal can also be used to create a convincing simulation of a "Kojak" police siren (typical of American police cars) using the FORMANT system. To do this, set the LFO frequency to approximately 1.5–2.5 Hz and feed the signal into the FM input of a VCO tuned to roughly 600 Hz, with the FM control set to approximately the midpoint. Additionally, using the keyboard's portamento function allows you to recreate the "Doppler effect" of a passing police car.

The sine signal is also highly useful for instrumental effects (organ, vibraphone, "musical saw," etc.). Finally, it can be used to generate the "sine percussion" sound familiar from electronic organs.

Parts List For Figure 1

Resistors (carbon film, 5%):

R1 = 120Ω
R2 = 2.7kΩ
R3, R11 = 10kΩ
R4, R7 = 18kΩ
R5 = 27kΩ
R6 = 1.5kΩ
R8, R9 = 12kΩ
R10 = 15kΩ
R12 = 33kΩ

Potentiometers:

P1 = 1kΩ (trimmer)
P2 = 4.7kΩ (trimmer)

Semiconductors:

IC1, IC2 = uA 741C (Mini-DIP)
D1 ... D10 = DUS (selected), e.g., 1N4148

Miscellaneous:

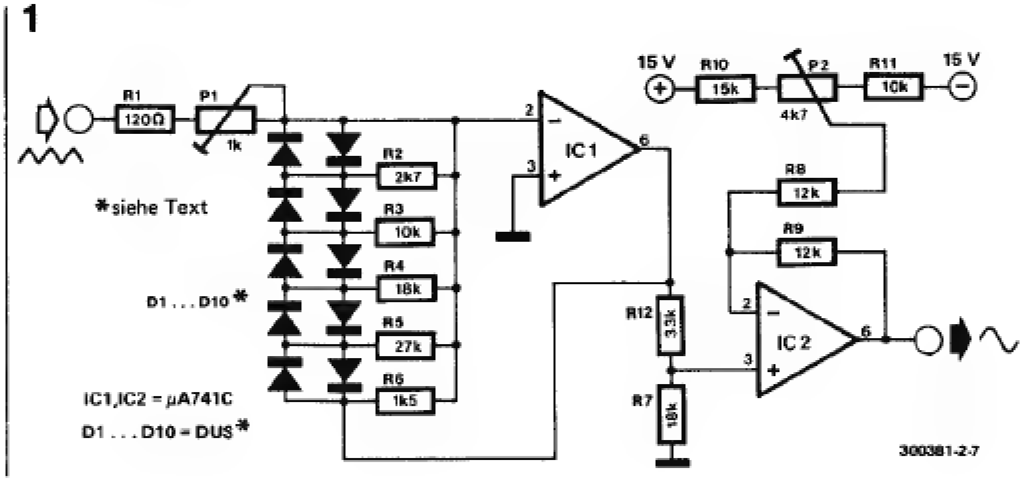
2 spacer sleeves (see text)

Figures

- Figure 1. Schematic diagram of the triangle-to-sine converter for the FORMANT LFOs.
- Figure 2. PCB layout and component placement plan for the circuit shown in Figure 1.
- Figure 3. Suggested mounting arrangement for the triangle-to-sine converter's add-on board relative to the FORMANT LFO board.

- Figure 4. The plotted waveforms serve as adjustment references. a) Adjustment and signal level are correct. b) Signal level is too low. c) The circuit is overdriven. d) The diodes are poorly matched.

1



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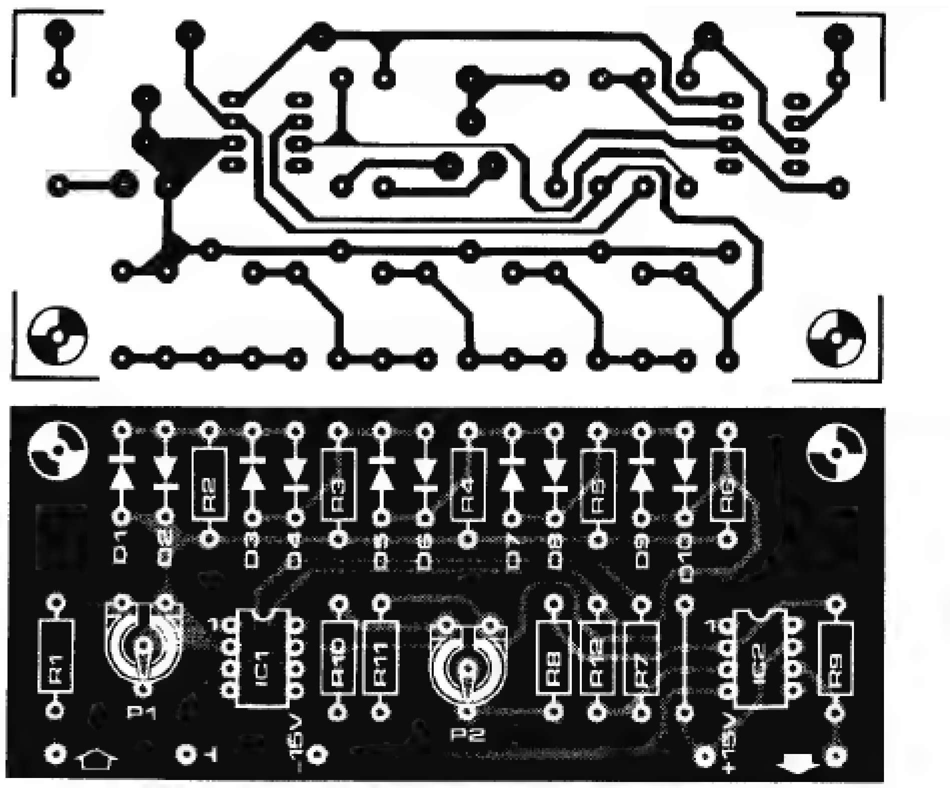

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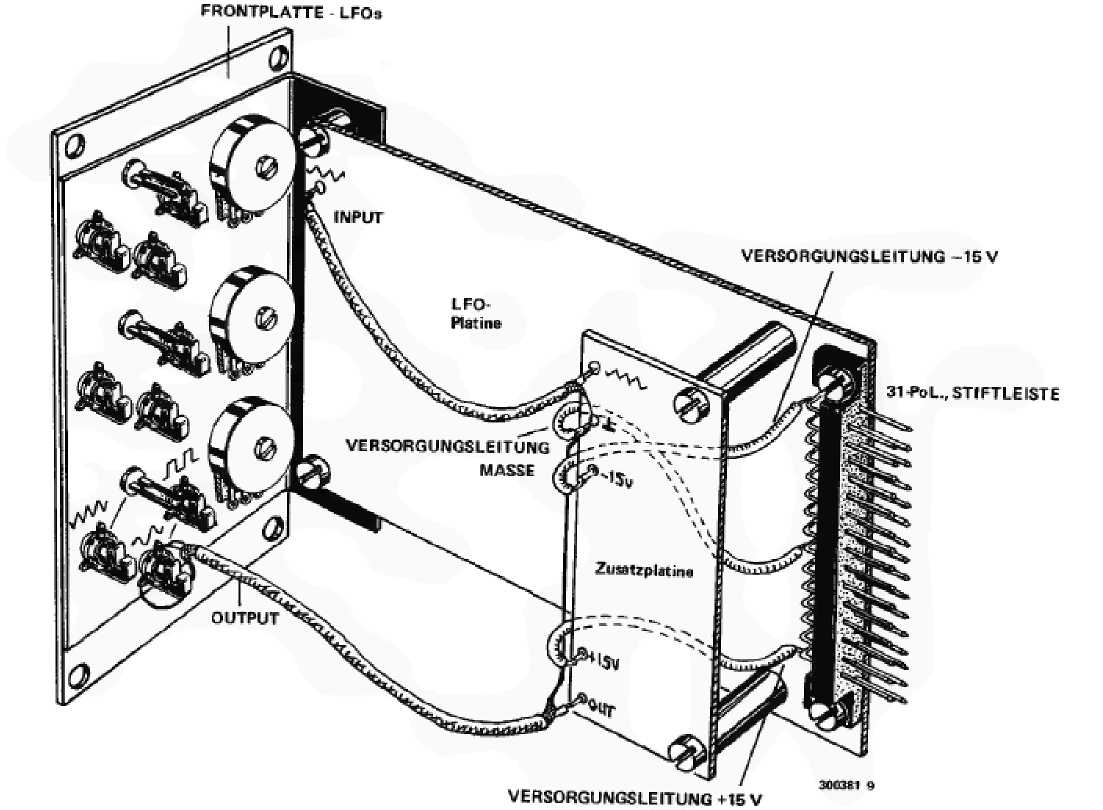
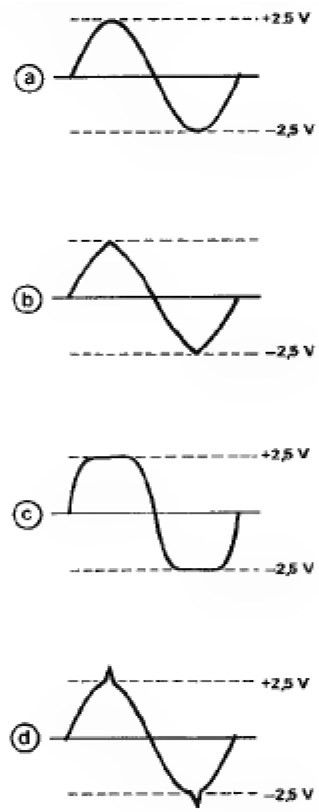

FRONTPLATTE - LFOs
INPUT
LFO-
Platine
VERSORGUNGSLEITUNG –15 V
31-PoL., STIFTLEISTE
VERSORGUNGSLEITUNG
MASSE
–15v
Zusatzplatine
OUTPUT
+15V
OUT
300381 9
VERSORGUNGSLEITUNG +15 V

4


+2,5 V
a
–2,5 V
b
–2,5 V
+2,5 V
c
–2,5 V
+2,5 V
d
–2,5 V